

Lab 14: OpenLDAP Implementation

Lab 14: OpenLDAP Implementation

This article will walk you through how to setup a simple OpenLDAP implementation with a Docker container. Within this article are all the steps needed to complete the lab, including the commands and scripts necessary, with screenshots inserted to help you follow along with the lab yourself.

Author

Joshua Draper

draperj2

VM Snapshot(s)

Virtual Machine 1542 (CIT371-001-Fall2025-draperj2-Ubuntu22Server) on node 'COIVMHOST2'

cit371-001-fall2025

▶ Start

🔌 Shutdown

📄 Summary

>_ Console

🖥 Hardware

☁ Cloud-Init

⚙ Options

📋 Task History

🕒 Snapshots

🛡 Firewall ▶

Take Snapshot | Rollback | Edit | Remove

Name	RAM	Date/Status	Description
🕒 Day1	No	2025-08-07 10:07:19	Used to revert VM to initial state. Do not delete!
└─ 🕒 Lab2	No	2025-09-04 12:44:29	Lab2
└─ 🕒 Lab3	No	2025-09-12 17:34:53	Lab3
└─ 🕒 Lab4	No	2025-09-13 11:00:54	Lab 4
└─ 🕒 Lab6	No	2025-10-01 13:53:21	Lab 6
└─ 🕒 Lab7	No	2025-10-08 13:48:03	Lab 7
└─ 🕒 Lab8	No	2025-10-09 17:22:43	Lab 8
└─ 🕒 Lab10	No	2025-10-16 16:35:56	Lab 10
└─ 🕒 Lab11	No	2025-10-23 15:56:47	Lab 11
└─ 🕒 Lab12	No	2025-11-04 09:32:31	Lab 12
└─ 🕒 Lab13	No	2025-11-14 17:28:41	Lab 13
└─ 🕒 Lab14	No	2025-11-21 16:12:28	Lab 14
🖥 NOW			You are here!

Overview

The purpose of LDAP is to provide a free, open-source, and highly customizable implementation of the Lightweight Directory Access Protocol. This can be used to build and manage directory services. It can be used to centrally store and manage information for authentication, authorization, and searching. This makes it the ideal tool for managing networks, their resources, and user data. OpenLDAP can be advantageous for businesses because it allows them to save money, since it is free open-source software, it helps them save time because it allows you to quickly and easily create and manage your own directory. It also offers improved security, protecting your catalog and data from various threats.

To compare OpenLDAP to Windows' Active Directory, Active Directory should be used if your environment is heavily Microsoft based; however, if you need a flexible cross-platform solution for a mixed use, opting for OpenLDAP is probably the better option. OpenLDAP is also free and open-source, whereas Active Directory is not. On the other hand, Active Directory does offer a nice GUI while OpenLDAP is primarily a command-line tool. Ultimately, if you are dealing with primarily Microsoft based hardware and software, it would be best to use Active Directory. If you need a flexible cross-platform tool to manage multiple systems at once, while doing it cost effectively, opt for OpenLDAP.

Procedure

The first step in this lab is to sign in to the Proxmox cluster and power on the PFSense and Ubuntu Server VM's. Then sign in to the Ubuntu Server VM.

Part 2: Setting up OpenLDAP

Open a terminal and enter the following command:

```
mkdir openldap
```

```
cd openldap
```

Then run this command to open the file:

```
sudo nano docker-compose.yaml
```

Next, copy and paste this information:

version: '2'

services:

ldap:

image: osixia/openldap:1.5.0

container_name: ldap

environment:

- LDAP_ORGANIZATION=371

- LDAP_DOMAIN=371.com

- "LDAP_BASE_DN=dc=371,dc=com"

- LDAP_ADMIN_PASSWORD=password

ports:

- 389:389

- 636:636

Your file should look like this:

A screenshot of a terminal window with a dark purple background. At the top, it shows 'GNU nano 6.2' and 'docker-compose.yml'. The content of the file is as follows:

```
version: '2'
services:
  ldap:
    image: osixia/openldap:1.5.0
    container_name: ldap
    environment:
      - LDAP_ORGANIZATION=371
      - LDAP_DOMAIN=371.com
      - "LDAP_BASE_DN=dc=371,dc=com"
      - LDAP_ADMIN_PASSWORD=password
    ports:
      - 389:389
      - 636:636
```

To save the file press Ctrl+O, Enter, Ctrl+X.

Enter the following command to run the container:

```
sudo docker compose up -d
```

Now run this command:

```
docker exec -it ldap /bin/bash
```

Copy and paste this information into the terminal and press Enter:

```
ldapadd -x -w password -D "cn=admin,dc=371,dc=com" << EOF
# LDIF file to add organizational unit "ou=devops" under "dc=371,dc=com"
dn: ou=devops,dc=371,dc=com
objectClass: organizationalUnit
ou: devops

# LDIF file to add organizational unit "ou=appdev" under "dc=371,dc=com"
dn: ou=appdev,dc=371,dc=com
objectClass: organizationalUnit
ou: appdev
EOF
```

Next, copy and paste this information and press Enter:

```
ldapadd -x -w password -D "cn=admin,dc=371,dc=com" << EOF
# LDIF file to Create user "amrutha" in "ou=appdev" under "dc=371,dc=com"
dn: cn=amrutha,ou=devops,dc=371,dc=com
objectClass: inetOrgPerson
cn: amrutha
```

sn: Amrutha
uid: amrutha
userPassword: Amrutha@123

LDIF file to Create user "charlie" in "ou=appdev" under "dc=371,dc=com"
dn: cn=charlie,ou=appdev,dc=371,dc=com
objectClass: inetOrgPerson
cn: charlie
sn: Charlie
uid: charlie
userPassword: Charlie@123

LDIF file to Create user "amit" in "ou=appdev" under "dc=371,dc=com"
dn: cn=amit,ou=appdev,dc=371,dc=com
objectClass: inetOrgPerson
cn: amit
sn: Amit
uid: amit
userPassword: Amit@123
EOF

Then copy and paste this information:

ldapadd -x -w password -D "cn=admin,dc=371,dc=com" << EOF
Group: appdev-team
dn: cn=appdev-team,dc=371,dc=com
objectClass: top
objectClass: groupOfNames
cn: appdev-team
description: App Development Team
member: cn=amrutha,ou=devops,dc=371,dc=com
member: cn=charlie,ou=appdev,dc=371,dc=com

Group: devops-team
dn: cn=devops-team,dc=371,dc=com
objectClass: top
objectClass: groupOfNames
cn: devops-team
description: DevOps Team
member: cn=amrutha,ou=devops,dc=371,dc=com
member: cn=amit,ou=appdev,dc=371,dc=com
EOF

Then copy and paste this one:

ldapadd -x -w password -D "cn=admin,dc=371,dc=com" << EOF
dn: cn=amrutha,ou=devops,dc=371,dc=com
changetype: modify

```
add: memberOf
memberOf: cn=devops-team,dc=371,dc=com
```

```
dn: cn=amit,ou=appdev,dc=371,dc=com
changetype: modify
add: memberOf
memberOf: cn=appdev-team,dc=371,dc=com
```

```
dn: cn=charlie,ou=appdev,dc=371,dc=com
changetype: modify
add: memberOf
memberOf: cn=devops-team,dc=371,dc=com
EOF
```

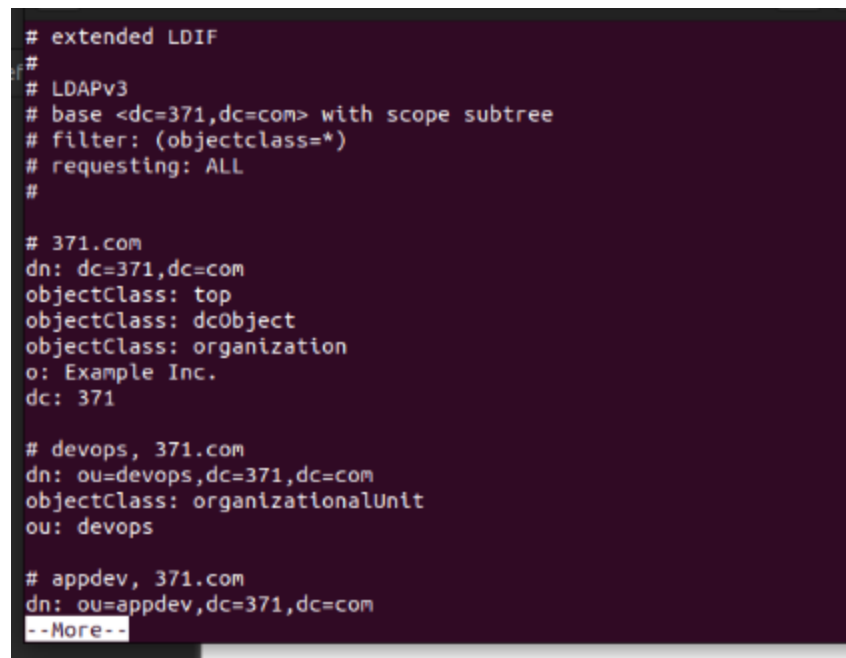
Part 3: Running LDAP Queries

Now, run this command to list everything you created:

```
ldapsearch -x -w password -D "cn=admin,dc=371,dc=com" -b "dc=371,dc=com" | more
```

You can press the Space bar to move down and see more information.

Your output should look like this:



```
# extended LDIF
#
# LDAPv3
# base <dc=371,dc=com> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# 371.com
dn: dc=371,dc=com
objectClass: top
objectClass: dcObject
objectClass: organization
o: Example Inc.
dc: 371

# devops, 371.com
dn: ou=devops,dc=371,dc=com
objectClass: organizationalUnit
ou: devops

# appdev, 371.com
dn: ou=appdev,dc=371,dc=com
--More--
```

Now run this command:

```
ldapsearch -x -w password -D "cn=admin,dc=371,dc=com" -b "dc=371,dc=com" "(cn=amit)"
```

Your output should look like this:

```

c=371,dc=com" "(cn=amit)"
# extended LDIF
#
# LDAPv3
# base <dc=371,dc=com> with scope subtree
# filter: (cn=amit)
# requesting: ALL
#
# amit, appdev, 371.com
dn: cn=amit,ou=appdev,dc=371,dc=com
objectClass: inetOrgPerson
cn: amit
sn: Amit
uid: amit
userPassword:: QW1pdEAXMjM=

# search result
search: 2
result: 0 Success

# numResponses: 2
# numEntries: 1
root@d64e505ad4bd:/#

```

Next, run this command:

```
ldapsearch -x -w password -D "cn=admin,dc=371,dc=com" -b "dc=371,dc=com" "(!((cn=appdev-team)(cn=devops-team)))"
```

Your output should look like this:

```

dn: cn=appdev-team,dc=371,dc=com
objectClass: top
objectClass: groupOfNames
cn: appdev-team
description: App Development Team
member: cn=amrutha,ou=devops,dc=371,dc=com
member: cn=charlie,ou=appdev,dc=371,dc=com

# devops-team, 371.com
dn: cn=devops-team,dc=371,dc=com
objectClass: top
objectClass: groupOfNames
cn: devops-team
description: DevOps Team
member: cn=amrutha,ou=devops,dc=371,dc=com
member: cn=amit,ou=appdev,dc=371,dc=com

# search result
search: 2
result: 0 Success

# numResponses: 3
# numEntries: 2
root@d64e505ad4bd:/#

```

Then run this command:

```
ldapsearch -x -w password -D "cn=admin,dc=371,dc=com" -b "dc=371,dc=com" "(objectClass=inetOrgPerson)"
```

Your output should look like this:

```
# charlie, appdev, 371.com
dn: cn=charlie,ou=appdev,dc=371,dc=com
objectClass: inetOrgPerson
cn: charlie
sn: Charlie
uid: charlie
userPassword:: Q2hhcmxpZUAXMjM=

# amit, appdev, 371.com
dn: cn=amit,ou=appdev,dc=371,dc=com
objectClass: inetOrgPerson
cn: amit
sn: Amit
uid: amit
userPassword:: QW1pdEAXMjM=

# search result
search: 2
result: 0 Success

# numResponses: 4
# numEntries: 3
root@d64e505ad4bd: /#
```

After you have run all of these commands you can exit the terminal, power off your VM's and sign out of the Proxmox cluster. You have successfully completed the lab!

References

MivoCloud:

[mivocloud](#) 

JumpCloud:

[LDAP vs Active Directory Vs Open LDAP: What's The Difference?](#) 

JumpCloud:

[Introduction to OpenLDAP - JumpCloud](#) 